

CHAPTER 1

INTRODUCTION

1.1 Background

Coronary Artery Disease (CAD) is the leading cause of death globally [4] and poses a significant public health challenge in India, with an increasing incidence at younger ages and higher mortality rates compared to global averages [4]. The disease results from the narrowing or blockage of coronary arteries [12], leading to impaired blood flow to the heart, which can cause heart attacks and other serious cardiovascular events. Early identification of individuals at risk of CAD is critical to implementing preventive measures that can reduce morbidity and mortality [1].

Risk prediction plays a vital role in CAD prevention by enabling clinicians to stratify patients according to their likelihood of developing the disease, allowing targeted interventions [9]. Currently, widely used clinical risk models predominantly rely on demographic, lifestyle, and traditional clinical factors such as age, gender, blood pressure, cholesterol levels, and smoking status [1]. While these models are useful at a population level, they do not incorporate the genetic predisposition that significantly contributes to an individual's risk [11,13], leading to potential underestimation of risk in some people [2].

1.2 Motivation for the study

Despite advances in cardiovascular risk prediction, existing tools have limitations [5]. Standard models frequently miss individuals with high inherited genetic risk due to a lack of integration of genomic data [1,13]. Research now suggests that genetics can account for 40% to 60% of CAD susceptibility [14], but this knowledge has yet to be fully harnessed in clinical practice [3].

Polygenic risk scores (PRS) have emerged as powerful tools that sum the effects of thousands of genetic variants to provide a comprehensive measure of inherited risk [1,2]. PRSs can improve discrimination of high-risk individuals beyond traditional risk factors, particularly in younger populations and those without overt clinical symptoms [7,9]. Furthermore, digital twin technology, which constructs a virtual, dynamic model of a patient based on longitudinal clinical and genetic information, presents an innovative approach to disease monitoring and risk prediction [6,8,15]. However, despite the promise of digital twins, there is limited clinical deployment [6] and a lack of integrated platforms that combine PRS with real-time health data for individualized risk prediction [10,12].

These gaps present an opportunity to develop integrated solutions that leverage advances in genomics and digital health for more precise, personalized CAD risk assessment and prevention [5].

1.3 Objectives of the project

The primary aim of this project is to develop and validate a translational digital twin platform that integrates polygenic risk scores with dynamic clinical data for real-time, individualized prediction of Coronary Artery Disease risk [6,10]. By combining genomic insights with ongoing clinical measurements, the platform seeks to provide a more comprehensive and accurate risk profile for patients [2,5].

Specific objectives include:

- Constructing a robust polygenic risk score tailored to the study population

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- Developing a dynamic digital twin model that incorporates both static genetic information and time-varying clinical data
 - Evaluating the predictive accuracy of the integrated platform compared to existing risk calculators
 - Demonstrating the utility of this approach for personalized preventive interventions in clinical workflows

Through this project, precision prevention of CAD can be advanced by early identification of high-risk individuals and continuous monitoring to inform timely, personalized clinical decision-making .

CHAPTER 2

PROBLEM STATEMENT

2.1 Current Challenges in CAD Risk Prediction

Current clinical risk prediction models for Coronary Artery Disease (CAD), including widely used calculators like QRISK3 and the ASCVD Pooled Cohort Equations (PCE), primarily rely on traditional clinical and lifestyle factors such as age, sex, blood pressure, cholesterol levels, diabetes status, and smoking history. While these models are effective at population-level risk stratification, they inadequately capture genetic predisposition, which is a substantial contributor to CAD susceptibility. Genetic factors are estimated to account for approximately 40% to 60% of CAD risk, yet most clinical calculators do not incorporate this data. This omission leads to under-recognition of high-risk individuals, particularly younger adults or those without conventional risk factors, thus missing opportunities for early preventive intervention [1].

Moreover, family history is often used as a surrogate for inherited risk but is an imprecise and incomplete marker, being influenced by environmental and lifestyle factors. The advent of polygenic risk scores (PRS), which aggregate the small effects of thousands of genetic variants, offers a promising avenue to fill this gap. However, despite their potential, PRSs are not routinely integrated into clinical workflows due to challenges, including lack of standardized platforms, uncertainty about how best to integrate genomic and clinical data, and limited accessibility.

Digital twin technology, emerging as a dynamic means to create patient-specific virtual models using continuous clinical data, holds promise for real-time disease management and prediction. However, digital twin approaches applied to CAD rarely incorporate genetic information, limiting their predictive power and personalization capability.

Together, these limitations highlight a critical challenge: the absence of an integrated, scalable framework that combines static genetic risk and dynamic clinical data to enable personalized, precise CAD risk prediction and management across diverse populations [2].

2.2 Need for Improved Risk Stratification

Improved risk stratification is essential to identify individuals at high risk of CAD who would benefit from early, targeted preventive measures such as lifestyle modification, pharmacotherapy, and monitoring. Conventional risk models can misclassify a significant subset of individuals, classifying genetically high-risk persons as low or intermediate risk due to a lack of genetic data incorporation. Such misclassification impedes the timely initiation of preventive interventions and could contribute to preventable cardiovascular events, including myocardial infarction and sudden cardiac death.

Recent multi-ethnic studies demonstrate that integrating PRS can reclassify up to 8% of borderline or intermediate-risk individuals, groups in which clinical decision-making is often ambiguous, into higher risk categories with up to a threefold increased CAD risk. This reclassification facilitates more precise clinical decision-making and resource utilisation [2]. Stratifying patients by combined clinical-genetic risk could thus substantially reduce disease burden and improve patient outcomes.

Furthermore, the dynamic nature of patient health necessitates tools that integrate evolving clinical parameters with genetic background to provide continuous risk updating, yet this remains an unmet need in current paradigms.

2.3 Specific Problem Addressed by the Project

This project addresses the critical unmet need for an integrated risk prediction platform that combines polygenic risk scores with time-resolved clinical data within a digital twin framework for real-time, individualized Coronary Artery Disease risk prediction. The specific problem lies in the absence of scalable, clinically deployable approaches that effectively fuse static genetic risk

measures with dynamic phenotypic and lifestyle information to improve predictive accuracy and clinical actionability.

By combining PRS with a digital twin model, this project aims to overcome the fragmented current landscape and develop a robust, translational predictive platform. This approach enables:

- Earlier and more accurate identification of high-risk individuals, especially those missed by traditional risk calculators
- Real-time monitoring of disease progression and risk factor changes within the personalized virtual patient model
- Improved precision prevention strategies by guiding timely clinical interventions based on integrated genomic and dynamic clinical risk assessment

The anticipated benefit is closing the translational gap between genetic research discoveries and routine CAD risk management, fostering personalized medicine and potentially reducing cardiovascular morbidity and mortality through better prediction and prevention [1][3][2].

CHAPTER 3

LITERATURE SURVEY

3.1 Sun, L., et al. (2021)

Title: *Polygenic risk scores in cardiovascular risk prediction. PLoS Medicine, 18(1), e1003498.*

This study systematically evaluates the clinical utility of polygenic risk scores (PRS) in improving cardiovascular disease (CVD) risk prediction beyond conventional methods. Utilizing data from over 300,000 participants in the UK Biobank, Sun et al. integrate PRS with established clinical calculators (incorporating variables such as age, cholesterol, and smoking status) to determine the effect on risk discrimination and classification. The inclusion of PRS leads to a statistically significant improvement in risk discrimination, evidenced by an increase in the C-index of 0.012. Notably, risk reclassification improves for both cases and non-cases.

A major finding is that targeted PRS testing among individuals at intermediate clinical risk can efficiently prevent additional CVD events. Their modeling suggests that one extra cardiovascular event can potentially be avoided for every 340 individuals screened at intermediate risk, a strategy far more efficient than population-wide PRS screening. The study also highlights that PRS outperforms biomarkers like C-reactive protein (CRP) in risk prediction, advancing risk prediction 1.5-fold compared to CRP-based approaches.

Strengths of this paper include its large sample size and robust statistical modeling, while limitations include potential lack of direct applicability to non-European ancestral groups represented in the UK Biobank. For your project, Sun et al.'s methodology and demonstration of PRS clinical value are directly relevant; their work validates the use of genetic information alongside clinical factors for

personalized risk stratification, which supports the development of integrated digital twin risk models for precision medicine.

3.2 Ratman, D., et al. (2025)

Title: *Polygenic risk scores improve CAD risk prediction in individuals at borderline and intermediate clinical risk. npj Cardiovascular Health, 1(1), pp. 13–25.*

Ratman et al. address the limitations of existing risk prediction algorithms for coronary artery disease (CAD), particularly for individuals categorized as borderline (5–7.5%) or intermediate (7.5–20%) risk by conventional clinical models. Their study builds and validates a polygenic risk score (PRS)-augmented model, caIRS, across four large, genetically diverse cohorts. The PRS integration yields consistent improvements in accuracy: a 1.5–3 percentage point increase in the C-index and net reclassification index gains up to 9.8%.

The paper shows that individuals with high PRS, previously classified as borderline or intermediate risk, actually possess a CAD hazard ratio up to 3.8 times higher than others, emphasizing the need for PRS inclusion in standard risk assessment. Ratman et al. also underscore the clinical significance of PRS for informing preventive treatments (like statin therapy) and for extending risk prediction accuracy to underrepresented ethnic populations.

The study's robust multi-ancestry validation is a notable strength, substantiating its utility across different demographic groups. However, there remain challenges in standardizing PRS for clinical application and ensuring real-world feasibility. This research is highly relevant for our project's goal to harmonize PRS with digital twins and dynamic patient data, demonstrating the necessity and technical feasibility of moving beyond static, phenotype-only models toward individualized care.

3.3 Marston, N.A., et al. (2023)

Title: *Predictive Utility of a Coronary Artery Disease Polygenic Risk Score for Statin Benefit. JAMA Cardiology, 8(6), pp. 692–700.*

Marston et al. investigate whether a polygenic risk score for CAD can enhance statin eligibility decisions, especially targeting younger adults who may be overlooked by existing clinical risk calculators. Analyzing a population of over 330,000, they show that individuals with high PRS have a substantially elevated risk for myocardial infarction, with the strongest prognostic utility observed in participants under age 50. The application of PRS among those with borderline or intermediate risk yields significant reclassification: 20% of individuals move up to a category warranting statin therapy, and 20% move down, indicating statins are unnecessary.

Key methods include prospective cohort analysis and statistical evaluation of PRS alongside demographic, biochemical, and clinical risk markers. The findings reveal that PRS-guided risk stratification is more effective in younger cohorts, who typically lack sufficient clinical risk accumulation for traditional calculators to warrant preventive treatment.

Strengths include large sample size and clinical relevance to medication decision-making; limitations are similar to other PRS papers, particularly regarding generalizability to non-European populations and real-world implementation. For our project, Marston et al.'s prospective approach and clear focus on clinical decision support through genetics offer strong support for integrating PRS into digital twin platforms for proactive, individualized care.

3.4 Iribarren, C., et al. (2024)

Title: *Polygenic risk and incident coronary heart disease in a multi-ethnic population. The Lancet Regional Health - Americas, 22, 100345.*

Iribarren and co-authors evaluate the predictive value of a 12-SNP polygenic risk score for incident coronary heart disease (CHD) across a multi-ethnic cohort (GERA) with up to 14 years of follow-up.

They demonstrate that PRS significantly improves long-term CHD risk stratification, particularly in Latinos, Europeans, and Asians, with more modest benefits seen in African-Americans. The study controls for conventional risk factors and family history, confirming that higher genetic risk quintiles are independently associated with increased CHD incidence.

Methods involve longitudinal cohort analysis with stratification by ethnic group, risk factor adjustment, and robust evaluation of predictive indices. The strength of this work lies in its real-world, long-term follow-up and demonstration of PRS reliability for lifetime risk assessment across diverse demographic groups. Limitations include varying effect sizes between ethnicities and the need for calibration or expansion of genetic algorithms for broader applicability.

This study reinforces the necessity of validating PRS-based models in multi-ethnic settings and the potential for tailored, population-level risk assessment strategies. For our project, Iribarren et al.'s findings underscore the importance of testing and calibrating integrated risk prediction tools for diverse ancestry groups to mitigate disparities and realize the promise of individualized, precision-based prevention.

3.5 Schunkert, H., et al. (2025)

Title: Clinical utility and implementation of polygenic risk scores for predicting cardiovascular disease

This paper reviews the evolving clinical applications and challenges of integrating polygenic risk scores (PRS) into cardiovascular disease (CVD) risk prediction frameworks. It emphasizes the growing evidence supporting PRS as an independent and additive predictor of CVD, complementing traditional risk factors. The authors discuss real-world implementation considerations including regulatory approval, standardization of PRS calculation, and integration into electronic health records.

Key highlights include the potential of PRS to refine risk stratification, particularly in intermediate-risk individuals who might benefit most from enhanced preventive strategies such as statin therapy. The review contextualizes PRS utility across diverse populations, underscoring the need for multi-ethnic validation and calibration to avoid exacerbating health disparities.

Importantly, the paper outlines barriers to clinical adoption such as clinician education, interpretability of genetic risk results, ethical concerns around genetic data use, and infrastructure needs for longitudinal data integration. The authors advocate for clinical guidelines and decision-support tools that incorporate PRS seamlessly into existing workflows.

For projects aiming to develop precision cardiovascular risk platforms or digital twin models, this study offers critical insights on clinical translation pathways, challenges to overcome for implementation, and the broader healthcare ecosystem's readiness for genomic-enhanced precision medicine. It substantiates the relevance of combining genomic data with clinical parameters to improve individualized risk prediction and preventive care.

3.6 Angelini, O., Christensen, N., Gude, M.F., & Bonar, S. (2025)

Title: *Digital twins: reimagining the future of cardiovascular risk prediction and personalised care*

This paper explores the emerging role of digital twin technology in cardiovascular disease (CVD) risk prediction and personalized healthcare. Digital twins are virtual replicas of patients that integrate multi-dimensional data—including genetics, clinical history, imaging, and real-time physiological measurements—into dynamic computational models that simulate disease progression and treatment responses.

The study highlights how digital twins enable continuous, personalized monitoring of cardiovascular health by modeling individual trajectories rather than relying on static risk assessments. By incorporating genomic data such as polygenic risk scores (PRS), digital twins provide a comprehensive framework linking genetic susceptibility with environmental and clinical factors.

Key contributions include demonstrations of how patient-specific digital twins can predict adverse cardiovascular events, optimize intervention timing, and forecast therapeutic outcomes. The paper also discusses technical challenges like data integration, model complexity, computational demands, and clinical translation barriers. Strategies to overcome these hurdles—such as advanced machine learning algorithms, interoperable health data systems, and clinician participation in validation—are examined.

This work underscores the transformative potential of digital twin frameworks and complements earlier discussions of PRS utility by situating genetic risk within a longitudinal patient context, advancing beyond static genomic risk models toward dynamic personalized care.

3.7 Park, Y.S., et al. (2024)

Title: *Evaluating cardiovascular disease risk stratification using multiple-polygenic risk scores and pooled cohort equations: insights from a 17-year longitudinal Korean cohort study*

This study examined whether combining multiple polygenic risk scores (multi-PRS) with the Pooled Cohort Equations (PCE) could improve cardiovascular disease (CVD) risk stratification in a Korean population. Using data from 7,612 genotyped adults in the Ansan and Ansung cohorts, the authors generated PRSs for myocardial infarction, ischemic stroke, coronary artery disease, and hypertension, and integrated them into a weighted composite score (wPRSsum). Participants were followed for up to 17 years.

The results showed that the wPRSsum was significantly associated with future CVD events, with individuals in the highest 20% of genetic risk showing elevated hazard ratios relative to low-risk groups. While adding the multi-PRS to the PCE produced only marginal changes in discrimination (C-index), combining both measures strengthened stratification, particularly in identifying subgroups with substantially elevated risk when both clinical and genetic risk were high.

This study demonstrates that ancestry-matched PRS can provide independent risk information in a non-European cohort and highlights how combining genetic and clinical data enhances personalized CVD risk prediction. Its findings are directly applicable to digital-twin frameworks that integrate stable genomic predisposition with dynamic clinical variables.

3.8 Coorey, G., Figtree, G.A., Fletcher, D.F., Snelson, V.J., Vernon, S.T., Winlaw, D., Grieve, S.M., McEwan, A., Yang, J.Y.H., Qian, P., O'Brien, K., Orchard, J., Kim, J., Patel, S., & Redfern, J. (2022)

Title: *The health digital twin to tackle cardiovascular disease – a review of an emerging interdisciplinary field*

This review examines the emerging concept of the health digital twin (HDT) and its growing relevance to cardiovascular disease (CVD). HDTs are virtual replicas of individuals created by integrating diverse data streams, including clinical, imaging, physiological, and other health data. The paper provides a structured mapping of 88 publications across multiple domains.

Most existing CVD digital twin work remains rooted in physics-based computational modelling, including electrophysiology simulations, haemodynamic models, and cardiac structural simulations. These systems operate as standalone tools and do not yet incorporate continuously updated, multimodal patient data.

The authors highlight that a true digital twin requires integrating real-time clinical, imaging, device, and biochemical data into adaptive models that evolve with the patient. They outline challenges—including data standardization, ethical considerations, and computational requirements—while emphasizing how HDTs can form the backbone of precision cardiovascular care.

The review's conclusions support projects aiming to combine genetic risk with longitudinal clinical trajectories, providing a foundation for building personalized CAD risk prediction systems that evolve over time.

3.9 Thangaraj, P.M., Benson, S.H., Oikonomou, E.K., Asselbergs, F.W., & Khera, R. (2024)

Title: *Cardiovascular care with digital twin technology in the era of generative artificial intelligence*

This review provides an overview of the expanding applications of digital twin technology in cardiovascular medicine, especially enhanced by recent AI and multimodal data advances. The digital twin is defined as an in silico model combining physiological, imaging, behavioral, and environmental data to replicate disease processes and forecast intervention outcomes.

The paper highlights use cases already entering clinical practice, such as virtual planning of coronary interventions, electrophysiology simulations for ablation planning, and structural heart models for procedural safety. It also covers emerging capabilities enabled by deep learning, including synthetic cardiac data generation, in silico trials, and integration of wearable/home-monitoring streams.

Despite these advances, the review identifies persistent barriers such as data sparsity, lack of interoperability, limited real-time updates, and regulatory challenges. These limitations must be addressed before digital twins can be scaled into routine care.

The review also connects genomic predisposition with digital twin systems, emphasizing that integrating lifelong genetic risks (e.g., PRS) with dynamic patient data can support more accurate and adaptive cardiovascular risk modeling.

3.10 Fuat, A., Adlen, E., Monane, M., Coll, R., Groves, S., Little, E., et al. (2024)

Title: *A polygenic risk score added to a QRISK2 cardiovascular disease risk calculator demonstrated robust clinical acceptance and utility in primary care*

This study assessed the clinical acceptance and utility of an integrated cardiovascular risk tool (CVD IRT), combining polygenic risk scores (PRS) with the QRISK2 calculator in UK primary care. The HEART study involved 832 individuals aged 45–64 years across twelve NHS practices.

The analysis demonstrated excellent technical feasibility, with valid reports generated for 100% of samples. Healthcare providers reported that the PRS-enhanced tool could be integrated seamlessly into routine workflows, and patients found the genomic information useful and easy to understand.

While average risk differed only modestly between QRISK2 and the combined tool, individual-level reclassification was significant: around 24% of participants shifted risk categories or experienced a $\geq 50\%$ change in predicted risk when PRS was added. Clinicians modified management decisions in 28% of such cases, rising to nearly 50% when risk increased by more than 2%.

This study provides strong real-world evidence that PRS can meaningfully alter risk perception and preventive strategies in everyday primary care. It supports integrating PRS as a lifelong genetic risk factor into personalized cardiovascular risk modelling systems.

CHAPTER 4

RESEARCH / TECHNOLOGY GAPS AND CHALLENGES

4.1 Research and Clinical Gaps

4.1.1 Limitations of Current Clinical Risk Predictors

The same rudimentary method has been in use for decades to determine who is at risk of heart attacks. Tools such as ASCVD-PCE and QRISK3, although widely used, rely only on traditional factors like age, cholesterol, blood pressure, smoking, and diabetes. However, research shows that a significant part of coronary artery disease (CAD) risk accumulates silently due to inherited genetic predisposition, long before clinical markers begin to change [14].

A typical case involves younger individuals with perfect clinical numbers who still develop premature CAD because their inherited risk remained unmeasured. Studies demonstrate that genetic predisposition contributes 40–60% of CAD risk, yet this dimension is absent in today’s calculators [14]. Family history is imprecise and incomplete, and high-risk individuals often remain unidentified.

Large-scale genomic studies show that polygenic risk scores (PRS) can identify 3–8% of the population with a threefold higher lifetime CAD risk—individuals who are routinely missed by clinical tools [11], [14]. Additionally, PRS performance varies across populations due to genetic diversity, making underdiagnosis even more likely in non-European groups [11], [13]. These blind spots highlight a major unmet need: incorporating ancestry-aware genomic risk into early prediction.

4.1.2 Barriers to Clinical Adoption of Genetic Risk Scores

Polygenic risk scores are scientifically well-established, with multiple large-scale studies confirming that individuals in the highest PRS tiers face 2–3× higher lifetime CAD risk even when biomarkers appear normal [14]. Multi-ancestry evaluations further show that 7–10% of individuals can be reclassified into more precise risk categories, improving prevention strategies [11], [13].

Despite this evidence, PRS adoption in real-world clinical practice remains extremely low. One reason is that clinicians often receive PRS as a single number without actionable context—there are no universally accepted guidelines on medication initiation, screening frequency, or intervention thresholds based on PRS [12].

Another limitation is that most PRS research treats risk as static, while real patients evolve over time. Lifestyle, comorbidities, medications, and aging all interact with genetic risk, yet current PRS studies rarely integrate these longitudinal changes [14], [15]. Moreover, genetic datasets used to develop most PRSs are overwhelmingly European, causing reduced accuracy—and sometimes misclassification—in underrepresented populations [11], [13].

These scientific and clinical gaps explain why PRS remains confined to research environments instead of becoming routine in preventive cardiology.

4.1.3 Digital Twin Models: Promise in Theory but Limited Practical Adoption

Digital twin technology—virtual patient models that simulate physiology, disease progression, and treatment response—is widely discussed in research but rarely used in real clinical settings. Current cardiovascular digital twin work remains limited to highly specialized models, such as blood-flow simulations or electrophysiology behavior, which do not capture the full complexity of a living patient [12], [15].

Creating a clinically useful digital twin requires integrating genetics, lifestyle, medications, imaging, physiological data, and continuous health updates. However, healthcare systems lack unified

longitudinal datasets with this breadth and quality [12]. Real-world EHRs, imaging, genomics, and wearable data are fragmented across platforms that cannot communicate.

Furthermore, many digital twin models remain computationally heavy, lacking real-time responsiveness and failing to integrate patient-specific molecular data like PRS [15]. Thus, most digital twin systems remain confined to laboratory prototypes rather than clinical decision-support tools.

4.1.4 Lack of Integrated Models Combining Genetic Risk and Patient Simulation

One of the most significant untouched opportunities in cardiovascular research is the integration of polygenic risk scores with dynamic patient simulations. Researchers working on genetic risk and those developing digital twins operate in separate domains, with limited attempts to merge the two into a unified system [12], [15].

A combined model could allow clinicians to visualize how a high-genetic-risk patient's CAD trajectory might evolve over ten years, and then simulate interventions such as statins, diet changes, exercise, or weight loss to see their impact in real time.

However, no platform currently integrates genetic predisposition into longitudinal heart disease simulations, despite evidence that genomic risk plays a major role across diverse populations [11], [13], [14]. Such a system would dramatically enhance preventive cardiology and address the gap where clinically “healthy-looking” individuals with high inherited risk remain undetected.

4.2 Technology Barricades and Infrastructure Problems

4.2.1 Fragmentation and Incompatibility of Healthcare Data

Healthcare generates vast amounts of heterogeneous data—genomics (VCF, PLINK), EHR (HL7, FHIR), imaging (DICOM), wearable data, and clinical notes—all in incompatible formats. Integrating these into a unified digital twin architecture requires extensive data harmonization, preprocessing, and standardization pipelines [12], [15].

Most research datasets are clean and structured, while real-world data is incomplete, noisy, and inconsistent. This discrepancy causes digital twin models built in controlled environments to perform poorly when deployed in real clinics.

Additionally, modern AI models powering digital twins remain largely “black boxes,” lacking interpretability. Clinical staff and regulators hesitate to trust risk predictions without a clear explanation of contributing factors, particularly when genomic inputs are involved [12].

4.2.2 Computational Demands and Barriers to Clinical Deployment

A clinically useful digital twin must update in near-real time when new patient data arrives. This requires high-performance computing (HPC), GPU clusters, scalable storage for genomic and imaging data, and low-latency architectures—resources unavailable to most hospitals and universities [15].

Even when prototypes perform well in labs, translating them into clinical environments involves regulatory approval, cybersecurity measures, interoperability with EHR systems, clinician interfaces, training, and multi-site validation—all of which form a massive translational gap [12], [15].

Thus, while digital twin technology shows promise, real-world deployment remains extremely challenging.

4.3 Data, Permissions, and Clinical Reality

4.3.1 Real-World Data Limitations and Population Biases

Genetic data often contains missing variants, imputation errors, and population biases, leading to reduced accuracy when PRS is applied to diverse groups [11], [13]. EHRs contain free-text notes, inconsistent coding, and missing follow-up information. Wearable data varies drastically by device and user behavior.

Longitudinal datasets are rarely complete because patients move, change hospitals, discontinue devices, or drop out of studies. Moreover, since most PRS models were built using European-majority datasets, applying them to Indian, African, or East Asian populations reduces precision significantly [11], [13].

These issues directly affect digital twin development, reducing predictive quality and increasing the risk of biased outcomes.

4.3.2 Regulatory, Ethical, and Consent Barriers

Accessing genetic and medical data requires navigating complex regulatory frameworks such as GDPR, HIPAA, and India's DPDP Act. Informed consent often does not cover future secondary usage, especially in digital twin modelling or genomic prediction research [12].

Genetic data also has heightened ethical sensitivity due to risks of discrimination, re-identification, and misuse. Digital twins add another layer of complexity because their predictions directly influence clinical decisions, generating questions of liability and fairness [15].

These practical, ethical, and legal barriers make large-scale data integration extremely difficult.

4.3.3 Clinical Acceptance and Patient Trust Challenges

Even the best models fail if clinicians don't adopt them. For adoption, tools must:

- improve patient outcomes,
- fit naturally into workflow,
- be interpretable, and
- be validated across diverse populations [12].

Patients also require transparency, privacy assurance, and trust, especially when genomic data is used. Without strong engagement from clinicians and patients, digital twin technology will remain experimental rather than practical.

CHAPTER 5

DATA EXPLORATION

5.1 Overview of Data Sources Across Studies

The reviewed studies are based on a number of large-scale cardiovascular research datasets. The UK Biobank is the most widely used resource, providing genomic data, clinical measurements, imaging, and long-term health records for more than 300,000 participants. Additional datasets include the GERA multi-ethnic cohort, which represents European, Latino, Asian, and African-American populations, and the Ansan-Ansung Korean Cohort, which supports longitudinal risk stratification in East Asian populations. Implementation-focused studies evaluating integrated genetic risk tools also incorporate real-world healthcare data, such as electronic health records from the UK NHS primary care system. Collectively, these datasets cover a broad range of populations, follow-up durations, and clinical settings.

5.2 Genomic Data and Polygenic Risk Scores

In almost all studies that use PRS, SNP genotyping arrays, and imputed variants are used to build polygenic risk scores based on genomic data. Such data will give a consistent, permanently valid estimate of familial predisposition to coronary artery disease (CAD) and other cardiovascular diseases. The multi-ancestry PRS validations of the GERA cohort and global consortia demonstrate that the effect sizes are dependent on the ancestry, yet genomic data should still be considered an important aspect of personalized risk prediction. Research also identifies the necessity of ancestry-congruent PRS calibration in order to provide equal performance.

5.3 Clinical and Lifestyle Risk Factor Data

All reviewed risk prediction models incorporate traditional cardiovascular risk factors as their foundational components. These include age, sex, blood pressure, lipid profiles, diabetes biomarkers, smoking status, body mass index, and medication use. These variables are sourced from national health databases (e.g., QRISK2/QRISK3), large research cohorts (such as the UK Biobank), or structured clinical assessments within NHS primary care settings. Such clinical and lifestyle data complement existing risk calculators and represent the dynamic elements that change over time—an essential requirement for digital twin systems that must continuously update predictions as a patient’s clinical status evolves.

5.4 Imaging and Physiological Data

Imaging records of coronary CT angiography, cardiac MRI, echocardiography, and electrophysiology are important for digital twin oriented studies. These data offer structural and functional data that are needed to construct patient specific heart models. Physiological measurements, such as ECGs, cardiac rhythms and haemodynamic measurements, are equally relevant to cardiac functionality simulation. Though imaging is less conspicuous in PRS-based work, it is core with regards to digital twins development since it enables mechanistic and statistical models to mimic individual cardiac behaviour.

5.5 Key Data Features for Digital Twin Integration

Several features across these datasets support digital twin development. Genomic data serve as fixed, high-value inputs that define baseline risk. Clinical and lifestyle variables provide modifiable risk factors that evolve with time. Imaging and physiological data offer the structural and functional basis

for computational modelling. Longitudinal and wearable data streams enable real-time or near-real-time updating. Together, these elements provide multimodal information needed to construct and maintain virtual patient models that reflect biological, behavioural, and environmental influences.

5.6 Data Challenges and Integration Barriers

Despite the richness of these datasets, several challenges emerge. Data heterogeneity across studies, including differences in measurement methods, coding systems, and follow-up, can complicate harmonization. Interoperability issues arise when combining genomic, imaging, and wearable data into a single analytical framework. Population diversity gaps, especially underrepresentation of minority groups, limit model fairness and generalizability. Variability in data quality, missing information, and differences in healthcare documentation practices also affect reliability. For digital twin applications, achieving seamless integration across these heterogeneous datasets remains a significant technical and organisational challenge.

5.7 Role of Combined Datasets in CAD Digital Twin Modelling

Taken together, these datasets provide the essential foundation for developing real-time, genetically informed CAD risk prediction tools. Genomic data offer stable inherited risk; clinical records and biomarkers track modifiable risk; imaging data enable precise modelling of cardiac structure and function; and wearable/EHR data support continuous monitoring. Integrating these elements allows digital twin models to simulate disease progression, forecast risk dynamically, and personalised

prevention strategies. The collective evidence highlights the feasibility—and the necessity—of combining multi-modal data sources to advance precision cardiovascular care.

CHAPTER 6

CONCLUSION OF CAPSTONE PROJECT PHASE - 1 AND FUTURE WORK

6.1 Conclusion

Phase-1 of the capstone project focused on establishing the scientific and conceptual groundwork for a digital-twin-enabled cardiovascular risk prediction framework. The literature review demonstrated strong evidence supporting the role of polygenic risk scores (PRS) in enhancing coronary artery disease (CAD) risk prediction, especially among individuals with borderline or intermediate clinical risk. Studies also showed growing feasibility of integrating PRS into routine primary care and highlighted the value of multimodal datasets—including genomic, clinical, imaging, and longitudinal health records—for personalised cardiovascular modelling.

A major outcome of Phase-1 was gaining clarity on the type of digital twin (DT) model appropriate for our project, as this was a key concern raised during Review-1. We initially explored two DT directions:

CFD-based mechanistic digital twin (using coronary CT angiography, 3D anatomy, haemodynamic simulation), and

Longitudinal clinical digital twin (using PRS, clinical variables, periodic measurements, and health records to model individual disease trajectories).

Through the Phase-1 analysis, it became evident that a full CFD digital twin requires high-resolution imaging, computational fluid dynamics pipelines, and structural heart modelling—resources not aligned with our current scope, available data, or project timeline. In contrast, the longitudinal,

data-driven digital twin aligns directly with our use of PRS, clinical risk calculators, and evolving patient data. This model type is supported by multiple studies demonstrating how genetic and clinical variables can be integrated to create adaptive, personalised risk prediction tools without requiring heavy biomechanical simulations.

Therefore, Phase-1 established that our project will adopt a longitudinal digital-twin framework, focusing on dynamic CAD risk forecasting using PRS alongside modifiable clinical and behavioural parameters.

6.2 Future Work

6.2.1 Development of the Longitudinal Digital Twin Architecture

The next phase involves constructing the foundational architecture of the longitudinal digital twin. This will include selecting appropriate temporal modelling strategies—such as recurrent neural networks, survival models, or hybrid state-space frameworks—to enable continuous and adaptive CAD risk prediction as new patient data becomes available. The architecture will incorporate modules for temporal data ingestion, periodic recalibration, and trend analysis to reflect evolving clinical states.

6.2.2 Integration of Polygenic Risk Scores into Dynamic Predictive Models

Future work will focus on embedding the PRS as a static, lifelong genetic anchor within the digital twin. Algorithms must be developed to fuse static genomic risk with modifiable clinical factors in a mathematically coherent manner. This will involve exploring weighting schemes, feature-fusion pipelines, and ancestry-specific calibration strategies to ensure PRS contributes meaningfully across diverse populations.

6.2.3 Construction or Acquisition of a Multimodal Dataset

Operationalizing the digital twin will require a representative dataset that captures PRS, demographic variables, classical CAD risk factors, and longitudinal follow-up data. Dataset development in the next phase may involve combining publicly available cohort data, generating synthetic patient records based on validated distributions, or using de-identified institutional datasets where permitted. Data preprocessing pipelines will be built for harmonization, missing-data handling, encoding of temporal variables, and multimodal fusion.

6.2.4 Implementation of Risk Forecasting and Intervention Simulation Modules

A key extension will be the development of simulation modules capable of projecting individualized CAD trajectories under different clinical or behavioural pathways. These modules will allow scenario-based forecasting—such as modelling the impact of lipid-lowering therapy, lifestyle modification, or changes in blood pressure—thereby enhancing the decision-support capabilities of the digital twin.

6.2.5 Development of PRS–Clinical Fusion Algorithms

Future efforts will include experimentation with advanced model architectures such as ensemble learning methods, gradient-boosted time-to-event models, and deep-learning pipelines designed for longitudinal prediction. The emphasis will be on improving discrimination and calibration while maintaining interpretability, robustness to missing data, and resistance to overfitting.

6.2.6 Evaluation of Predictive Performance, Interpretability, and Fairness

In subsequent phases, the digital twin will be evaluated using clinically relevant metrics such as the C-index, calibration plots, net reclassification improvement, and ancestry-stratified performance.

Model explainability will be incorporated using interpretability frameworks (e.g., SHAP-based feature attribution). Ensuring fairness and mitigating ancestry-based bias will be prioritized to support eventual clinical integration.

6.2.7 Development of a Clinician-Facing Visualisation and Decision-Support Interface

A prototype interface will be developed to present dynamic risk trajectories, intervention simulations, and tailored recommendations derived from the digital twin. The interface will prioritize clarity, interpretability, and clinical workflow compatibility, serving as an early model for user-centered digital twin deployment in clinical settings.

6.2.8 Exploration of Real-Time Wearable and Remote Monitoring Integration

Later phases may extend the digital twin by incorporating high-frequency data streams from wearables or home-monitoring devices (e.g., heart rate variability, activity, sleep metrics). Such integration would enable short-term risk tracking and more responsive updates to the digital twin, enhancing its temporal precision.

6.2.9 Ethical, Regulatory, and Implementation Considerations

As the system progresses toward translational readiness, the project will address ethical, regulatory, and privacy considerations surrounding genomic data use. Future work will examine pathways for compliance with DPDP, HIPAA, and GDPR, evaluate informed consent frameworks, and explore requirements for clinical deployment—including EHR interoperability, cybersecurity safeguards, and multi-site real-world validation

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APPENDIX A: DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

Term / Acronym	Definition / Description
CAD	Coronary Artery Disease
CVD	Cardiovascular Disease
PRS	Polygenic Risk Score – a score based on the cumulative effect of many genetic variants
SNP	Single Nucleotide Polymorphism – a variation at a single position in a DNA sequence
ASCVD	Atherosclerotic Cardiovascular Disease

ASCVD-PCE	ASCVD Pooled Cohort Equations – a risk calculator for cardiovascular events
QRISK2 / QRISK3	Risk calculators used to estimate the ten-year risk of CVD
PCE	Pooled Cohort Equations – standard tool for CVD risk estimation
EHR	Electronic Health Record
DT / Digital Twin	A virtual, dynamic model of a patient using data from clinical, imaging, and genomic sources
GERA	Genetic Epidemiology Research on Adult Health and Aging cohort
UK Biobank	A large-scale biomedical database in the UK containing genetic and health information
NHS	National Health Service (UK)

BMI	Body Mass Index
ECG	Electrocardiogram
CT	Computed Tomography
MRI	Magnetic Resonance Imaging
VCF	Variant Call Format – a file format for storing gene sequence variations
BGEN	File format for storing genetic data
PLINK	Software/package for whole-genome association analysis and data file format
HL7 / FHIR	Health Level Seven / Fast Healthcare Interoperability Resources – standards for exchanging healthcare information electronically
DICOM	Digital Imaging and Communications in Medicine – standard for handling medical images

CFD	Computational Fluid Dynamics – methods for simulating fluid flow, sometimes used in heart modelling
wPRSsum	Weighted Polygenic Risk Score summary – composite of multiple PRS scores
HDT	Health Digital Twin
AI	Artificial Intelligence
GDPR	General Data Protection Regulation (EU)
HIPAA	Health Insurance Portability and Accountability Act (US)
FDA	Food and Drug Administration (US regulatory body)
C-index	Concordance Index – a measure of predictive accuracy
Hazard Ratio	A measure of risk for a particular outcome in one group compared to another

Statin	A class of lipid-lowering medications for CVD prevention
Imputation	The statistical inference of missing genotype data in genetic studies
Longitudinal Data	Data collected over time from the same subjects
Multimodal Data	Data from multiple sources (e.g., clinical, genomic, imaging, wearable)